

Number Series (संख्या श्रृंखला)

Complete Theory, Solved Examples & Practice Set

1. Theory & Concepts (सिद्धांत)

A **Number Series** is a sequence of numbers that follows a specific mathematical pattern or logical rule. To solve these questions, you must identify the hidden rule governing the sequence and apply it to find the missing or next number.

Common Types of Number Series:

- **Arithmetic Series (जोड़/घटाव):** The difference between consecutive terms is constant or follows a pattern (e.g., +2, +4, +6 or -3, -3, -3).
- **Geometric Series (गुणा/भाग):** Each term is multiplied or divided by a specific number (e.g., $\times 2$, $\times 3$, $\times 4$).
- **Prime Number Series (अभाज्य संख्या):** Series based on prime numbers (2, 3, 5, 7, 11, 13...).
- **Perfect Square/Cube Series (वर्ग और घन):** The numbers are squares or cubes of integers, sometimes with a constant added or subtracted (e.g., n^2 , n^3 , $n^2 + 1$, $n^3 - 1$).
- **Alternating/Mixed Series (मिश्रित श्रृंखला):** Two different series are intertwined, or operations alternate (e.g., +2, $\times 3$, +2, $\times 3$).

Pro Tip: Always check the difference between consecutive numbers first. If the difference increases rapidly, the series likely involves multiplication or squares/cubes.

2. Solved Examples (उदाहरण और हल)

Example 1: 5, 10, 15, 20, 25, ?

Solution: The pattern is adding 5 to the previous number. ($25 + 5 = 30$)

Example 2: 3, 6, 12, 24, 48, ?

Solution: Each number is multiplied by 2. ($48 \times 2 = 96$)

Example 3: 2, 3, 5, 7, 11, ?

Solution: This is a continuous series of Prime Numbers. The next prime number after 11 is **13**.

Example 4: 1, 4, 9, 16, 25, ?

Solution: These are perfect squares: $1^2, 2^2, 3^2, 4^2, 5^2$. The next is $6^2 = 36$.

Example 5: 120, 99, 80, 63, 48, ?

Solution: The pattern is $n^2 - 1$. ($11^2-1, 10^2-1, 9^2-1, 8^2-1, 7^2-1$). Next is $6^2 - 1 = 36 - 1 = 35$.

Example 6: 2, 5, 10, 17, 26, ?

Solution: The pattern is $n^2 + 1$. ($1^2+1, 2^2+1, 3^2+1, 4^2+1, 5^2+1$). Next is $6^2 + 1 = 37$.

Example 7: 10, 14, 28, 32, 64, 68, ?

Solution: The alternating pattern is (+4) then ($\times 2$). ($10+4=14, 14 \times 2=28, 28+4=32, 32 \times 2=64, 64+4=68$). Next is $68 \times 2 = 136$.

Example 8: 50, 49, 46, 41, 34, ?

Solution: The differences are consecutive odd numbers: -1, -3, -5, -7. Next difference is -9. ($34 - 9 = 25$).

Example 9: 2, 4, 12, 48, 240, ?

Solution: Multiplication by consecutive numbers: $\times 2, \times 3, \times 4, \times 5$. Next is $\times 6$. ($240 \times 6 = 1440$).

Example 10: 5, 11, 23, 47, 95, ?

Solution: The pattern is $(\times 2 + 1)$. ($5 \times 2 + 1 = 11$, $11 \times 2 + 1 = 23$, $47 \times 2 + 1 = 95$). Next is $95 \times 2 + 1 = 191$.

Example 11: 3, 4, 8, 17, 33, ?

Solution: The differences are perfect squares: $+1^2, +2^2, +3^2, +4^2$ ($+1, +4, +9, +16$). Next difference is $+5^2$ ($+25$). ($33 + 25 = 58$).

Example 12: 1, 8, 27, 64, 125, ?

Solution: These are perfect cubes: $1^3, 2^3, 3^3, 4^3, 5^3$. The next is $6^3 = 216$.

3. Practice Questions (अभ्यास प्रश्न)

Q1. 6, 12, 18, 24, ?

(A) 28

(B) 30

(C) 32

(D) 34

Q2. 2, 4, 8, 16, ?

(A) 24

(B) 28

(C) 32

(D) 36

Q3. 11, 13, 17, 19, 23, ?

(A) 25

(B) 27

(C) 29

(D) 31

Q4. 1, 4, 9, 16, 25, ?

(A) 30

(B) 36

(C) 40

(D) 49

Q5. 120, 99, 80, 63, 48, ?

(A) 35

(B) 38

(C) 39

(D) 40

Q6. 5, 11, 23, 47, ?

(A) 93

(B) 94

(C) 95

(D) 96

Q7. 2, 7, 14, 23, 34, ?

(A) 45

(B) 47

(C) 49

(D) 50

Q8. 8, 27, 64, 125, ?

(A) 216

(B) 256

(C) 343

(D) 512

Q9. 7, 14, 42, 168, ?

(A) 672

(B) 840

(C) 756

(D) 920

Q10. 10, 12, 15, 20, 27, ?

(A) 35

(B) 36

(C) 37

(D) 38

Q11. 3, 8, 15, 24, 35, ?

(A) 46

(B) 48

(C) 50

(D) 52

Q12. 5, 25, 125, 625, ?

(A) 1250

(B) 2500

(C) 3125

(D) 4025

Q13. 40, 35, 30, 25, 20, ?

(A) 10

(B) 12

(C) 15

(D) 18

Q14. 4, 16, 36, 64, 100, ?

(A) 120

(B) 144

(C) 169

(D) 196

Q15. 1, 2, 6, 24, 120, ?

(A) 240

(B) 500

(C) 720

(D) 840

Q16. 100, 50, 25, 12.5, ?

(A) 5.00

(B) 6.00

(C) 6.25

(D) 6.50

Q17. 2, 3, 6, 18, 108, ?

(A) 1944

(B) 1608

(C) 1296

(D) 2106

Q18. 3, 6, 11, 18, 27, ?

(A) 36

(B) 38

(C) 40

(D) 42

Q19. 9, 28, 65, 126, ?

(A) 217

(B) 216

(C) 215

(D) 218

Q20. 8, 15, 28, 53, ?

(A) 100

(B) 102

(C) 104

(D) 106

Q21. 13, 14, 18, 27, 43, ?

(A) 58

(B) 68

(C) 78

(D) 88

Q22. 21, 25, 33, 49, 81, ?

(A) 113

(B) 129

(C) 145

(D) 161

Q23. 1, 5, 14, 30, 55, ?

(A) 85

(B) 90

(C) 91

(D) 95

Q24. 0, 3, 8, 15, 24, ?

(A) 33

(B) 35

(C) 37

(D) 39

Q25. 12, 12, 24, 72, 288, ?

(A) 1440

(B) 1200

(C) 1152

(D) 1360

4. Answer Key (उत्तरमाला)

Q1. (B)

Q2. (C)

Q3. (C)

Q4. (B)

Q5. (A)

Q6. (C)

Q7. (B)

Q8. (A)

Q9. (B)

Q10. (D)

Q11. (B)

Q12. (C)

Q13. (C)

Q14. (B)

Q15. (C)

Q16. (C)

Q17. (A)

Q18. (B)

Q19. (A)

Q20. (B)

Q21. (B)

Q22. (C)

Q23. (C)

Q24. (B)

Q25. (A)